National University of Computer and Emerging Sciences

Islamabad Campus

Numerical Computing (CS2008)

Date: Nov 8th 2025

Course Instructor(s)

Dr. Muhammad Rizwan

Section(s): BS(CS) A 2022

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 5

_____

Invigilator's Signature

_____ _____ _____

Roll No Section Student Signature

Do not write below this line

Attempt all the questions.

Question No	Q1	Q2	Q3	Q4	Q5	Total
Marks Obt.						
Max Marks	10	10	10	10	10	50

Attempt all questions on answer book.

Instructions:

Conduct & Materials

- Mobile phones and smart devices are not allowed.
- Cutting/erasing/overwriting is not allowed.
  Do all rough work on a separate sheet and keep your final answers clean.
- Only non-programmable calculators are allowed (unless explicitly permitted).
- Use blue/black ink; do not use red/green. Write legibly.

Answering Rules

- Number your answers clearly and in the same order as the question paper.
- Where a method is required, show essential steps; unsupported final answers may receive reduced credit.
- Units, labels, and diagrams must be clear and properly scaled where applicable.

Marking & Deductions

- Cuts/corrections/overwriting on final answers will result in mark deductions. (Neat, clean answers get full consideration.)
- Illegible or ambiguous answers may not be marked.

Optional Bonus Question

- There is one optional bonus question (only one part).
  - If you skip it, no marks are deducted.
  - If you solve it correctly, you earn bonus marks as indicated.

Academic Integrity

- This is an individual exam. Cheating, communicating, or sharing material leads to disqualification and disciplinary action.
- Do not remove question paper, answer sheets, or any exam materials without permission.

Submission

- Attach any rough sheets at the end (clearly labeled "Rough Work").
- Do not staple inside the main booklet, use the cover staples only.
- Stop writing when time is called, late additions will not be considered.
- Handover your question-and-answer booklet to the invigilator before leaving.

[CLO 1: Apply numerical methods for various mathematical operations…]

Q1:

a) Construct Newton's Divided Difference Table for the following data

x	1	2	4	7
f(x)	3	6	12	24

b) Use Newton's Forward Difference Formula to estimate f(1.5):

x	1.0	1.2	1.4	1.6
f(x)	2.0	2.303	2.639	3.0

[CLO 1: Apply numerical methods for various mathematical operations…]

Q2:

a) Explain Hermite Interpolation. Why do we include derivative values? Provide a simple example.

b) Describe Runge's Phenomenon and explain why splines help to avoid it.

c) For Quadratic Splines, how many unknowns and how many equations are generated for n+1 data points? Show with reasoning.

d) Explain Numerical Interpolation, Integration and Differentiation, when to use these?

[CLO 1: Apply numerical methods for various mathematical operations…]

Q3: Gaussian Quadrature Problem

a) Use 2-point Gaussian Quadrature to approximate:

$$\int_{-1}^{1} 5\, e^{x}\, dx$$

Then compare your result to the exact value.

b) Approximate

$$\int_{0}^{2} x^{2} \ln\,(x^{2}+1)\; dx$$

using $h = 0.25$ by applying composite trapezoidal rule

[CLO 1: Apply numerical methods for various mathematical operations…]

Q4:

a) Construct the quadratic spline that passes through the points $(1,2), (2,3), (3,5)$.

b) Let $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 6$ and $f(x) = e^x$. Use the Lagrange interpolating polynomial $p_{1,2,4}(x)$(i.e., nodes $x_1, x_2, x_4 = \{2,3,6\}$) to approximate $f(5)$.

[CLO 1: Apply numerical methods for various mathematical operations…]

Q5:

a) Use Simpson's $\frac{3}{8}$rule to evaluate

$$\int_{10}^{16} f(x)\ dx,$$

using the data:

x	10	11	12	13	14	15	16
$f(x)$	1.02	0.94	0.89	0.79	0.71	0.62	0.55

b) Use numerical differentiation to complete the table:

x	$f(x)$	$f'(x)$
1.1	9.02501	
1.2	11.02318	
1.3	13.46374	
1.4	16.44465	

Formulas

$$f[x_i] = f(x_i),$$

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i},$$

$$f[x_i, \ldots, x_{i+k}] = \frac{f[x_{i+1}, \ldots, x_{i+k}] - f[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i}$$

$$P_n(x) = f[x_0] + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \cdots$$

$$y = \sum_{i=0}^{n} y_i \left(\prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} \right)$$

$$f(x) = f_0 + r\,\Delta f_0 + \frac{r(r-1)}{2!}\Delta^2 f_0 + \ldots\ldots + \frac{r(r-1)\ldots\ldots(r-n-1)}{n!}\Delta^n f_0$$

$$r = \frac{x - x_j}{h}$$

Quadratic spline

For n+1 points, there will be n intervals and for each interval there will be a 2nd order polynomial

Total unknowns: 3n. Use the following equations:

- The first and last functions must pass through the end points (2 equations).
- The function values must be equal at interior points (2n–2 equations).
- First derivatives must be equal at the interior points (n–1 equations).
- These give (2 + (2n-2) + (n-1) = 3n-1) equations; one additional boundary condition is required to reach (3n) equations.

Common choices (pick one): a1 = 0

The polynomial is given by:

$$P_{2n+1}(x) = \sum_{i=0}^{n} A_i(x) f_i + \sum_{i=0}^{n} B_i(x) f_i'$$

where:

- $f_i = f(x_i)$
- $f_i' = f'(x_i)$

Definitions of Coefficients

$$A_i(x) = [1 - 2(x - x_i)L_i'(x_i)]\, L_i^2(x)$$

$$B_i(x) = (x - x_i)L_i^2(x)$$

Here, $L_i(x)$ are the **Lagrange basis polynomials** defined as:

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

$$I \approx \text{Midpoint Rule } (M_1) = (b - a) \cdot f\left(\frac{a + b}{2}\right)$$

$$I \approx \sum_{i=1}^{n} f(\bar{x}_i)\, \Delta x$$

$$I \approx \frac{b - a}{2}[f(a) + f(b)]$$

$$I \approx \frac{b - a}{2n}[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)]$$

$$I = \frac{h}{3}(f_0 + 4f_1 + f_2)$$

$$\frac{1}{3}h\left[f(x_0)+4\sum_{i=1}^{n/2}f(x_{2i-1})+2\sum_{i=1}^{n/2-1}f(x_{2i})+f(x_n)\right]$$

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0+3f_1+3f_2+f_3\right]$$

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f_0+3\sum_{\substack{1\le i\le n-1\\ i\not\equiv 0 \pmod 3}} f_i+2\sum_{\substack{3\le i\le n-3\\ i\equiv 0 \pmod 3}} f_i+f_n\right], \quad n=3m$$

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\sum_{i=1}^{N} w_i f\left(\frac{b-a}{2}t_i+\frac{a+b}{2}\right)$$

Number of points, n	Points, x_i		Weights, w_i	
1	0		2	
2	$\pm\frac{1}{\sqrt{3}}$	±0.57735...	1	
3	0		$\frac{8}{9}$	0.888889...
	$\pm\sqrt{\frac{3}{5}}$	±0.774597...	$\frac{5}{9}$	0.555556...
4	$\pm\sqrt{\frac{3}{7}-\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.339981...	$\frac{18+\sqrt{30}}{36}$	0.652145...
	$\pm\sqrt{\frac{3}{7}+\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.861136...	$\frac{18-\sqrt{30}}{36}$	0.347855...
5	0		$\frac{128}{225}$	0.568889...
	$\pm\frac{1}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}$	±0.538469...	$\frac{322+13\sqrt{70}}{900}$	0.478629...
	$\pm\frac{1}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}$	±0.90618...	$\frac{322-13\sqrt{70}}{900}$	0.236927...

$$f'(x_i)=\frac{f(x_{i+1})-f(x_i)}{h}+O(h)$$